Southern New Hampshire University

# CS 255 Business Requirements

## System Components and Design

### Purpose

*What is the purpose of this project? Who is the client and what do they want their system to be able to do?*

The purpose of this system design is to train students for their driving exam and on the road test at the Department of Motor Vehicles. This project was offered by the Owner of DriverPass. This system will come in three tiers of access for students. Services offered will be on the road training and practice exams. The system should have different user privileges to the Students and Administrators. The system should also alert Administrators of changes in DMV policy.

### System Background

*What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?*

The proposed DriverPass system will be designed to handle the needs of Students in scheduling drivers and completing exams. Students should also be able to choose from three packages, the first option being six hours in person driving, the second being eight hours driving and an in-person lesson on Department of Motor Vehicles rules and policies and finally the third tier would include twelve hours in person driving, an in person lesson and online class. The packages cannot be added to or removed but can be disabled by the Owner. The DriverPass application should have built in roles and security profiles. The Owner should have full administrative access over all accounts and have all privileges. The Information Technology Officer should have maintenance and necessary administrative privileges. The Secretary should be able to make and modify appointments. Finally, the Students should be able to make and modify appointments as well. Notifications of Department of Motor Vehicle changes in policy or rules should be sent to DriverPass.

### Objectives and Goals

*What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?*

- Package Management
  - Students are able to subscribe to one of the three packages
  - The owner should be able to disable a tier if it gets too full
  - Package one should be six hours with trainer
  - Package two should be eight hours trainer and in person lesson
  - Package three should be twelve hours trainer, in person lesson and online class
- Access Control and Security
  - Owner is the administrator with ability to give and revoke access and disable packages

    - Information Technology Officer has necessary privileges to maintain system
    - Secretary can schedule driving instruction
    - Students can schedule driving instruction and participate in online class
- System Infrastructure
    - System is fully hosted in the Cloud
    - Pc and mobile access to documents such as excel spreadsheets

## Requirements

### Nonfunctional Requirements

*In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.*

**Performance Requirements**

*What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?*

- Responsive Design
    - System will have zero-downtime deployment
    - The Department of Motor Vehicles change in policy could require update on the fly
- Cross-Platform Capability
    - Application will run in browser
    - System will support downloads such as spreadsheets
        - The system must be accessible by browser to meet the needs of the students and owner

**Platform Constraints**

*What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?*

- Compatibility
    - System will be platform independent
    - Students will not need to worry about which OS or browser they are using
        - This is convenient for students
        - Gives DriverPass Wider Reach
        - The system must be compatible with all students on all systems to maximize the reach and convenience of DriverPass

- Reliability
    - The system will be cloud based
    - Application should be stored and ran from a cloud service
        - Cloud services could be Google Cloud or Amazon Web Services
    - Cloud services have Scalability (*Why Partnership Matters*, 2026)
        - Cloud services scale on the fly
        - Pay as you go to scale economically

- High uptime
  - The system must have reliable and scalable systems to meet the current and future needs of DriverPass

**Accuracy and Precision**

*How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?*

- Security
  - Application will have roles and privileges
  - Application will have tracking of user actions
  - Application will have industry standard password protection for accounts
  - Students will be identified by appropriate package tier
    - The system must have appropriate security measures to prevent unauthorized access or actions so that the integrity of the application is not undermined
- Support
  - Users can recover accounts
  - Alerts concerning maintenance will be sent to Information Technology Officer
  - Alerts concerning Department of Motor Vehicles will be sent to Owner
    - System must alert appropriate users of events so that they can make take. Appropriate actions

**Adaptability**

*Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?*

- Dynamic Administration
  - Secretary can create and modify user accounts
  - Owner can grant and revoke privileges
  - Information Technology Officer can maintain system
  - Owner can disable packages
  - Students can reset password
  - Developers required for platform updates
    - The application must be able to adapt to changes on the fly so that the business functions are not interrupted from day to day

**Security**

*What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a "brute force" hacking attempt? What happens if the user forgets their password?*

- Authentication
  - Users will have passwords and optional 2 factor Authentication
  - Input validation and sanitization

- Will use Hypertext Transfer Protocol Secure (HTTPS) between client and server
- Cloud application will be redundant, so data is not lost in event of attack
- Data will be encrypted, so sensitive data is secured in event of attack
- Password reset can be done individually or with secretary

**Functional Requirements**

*Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."*

**User Interface**

*What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?*

- The system shall allow Students to:
  - subscribe to a package
  - schedule driving appointments
  - modify driving appointments
  - take online tests
  - recover account
  - use any browser or operating system
  - see online test progress
  - update information
  - list special needs
  - view driver notes
- The system shall allow Secretaries to:
  - Schedule appointments for students
  - Modify appointments for students
  - Assist in account recovery for students
- The system shall allow the Information Officer to:
  - Maintain system
  - Modify system
- The system shall allow the Owner to:
  - Grant and revoke privileges
  - Have full administrative control
  - Disable packages
  - Enable packages
  - Download reports

**Assumptions**

*What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?*

- Reliable internet access
    - It is assumed students and staff will have continuous and reliable internet access since DriverPass is hosted online
- Student Technology required
    - It is assumed students will have the platforms available to access the course using the internet

## Limitations

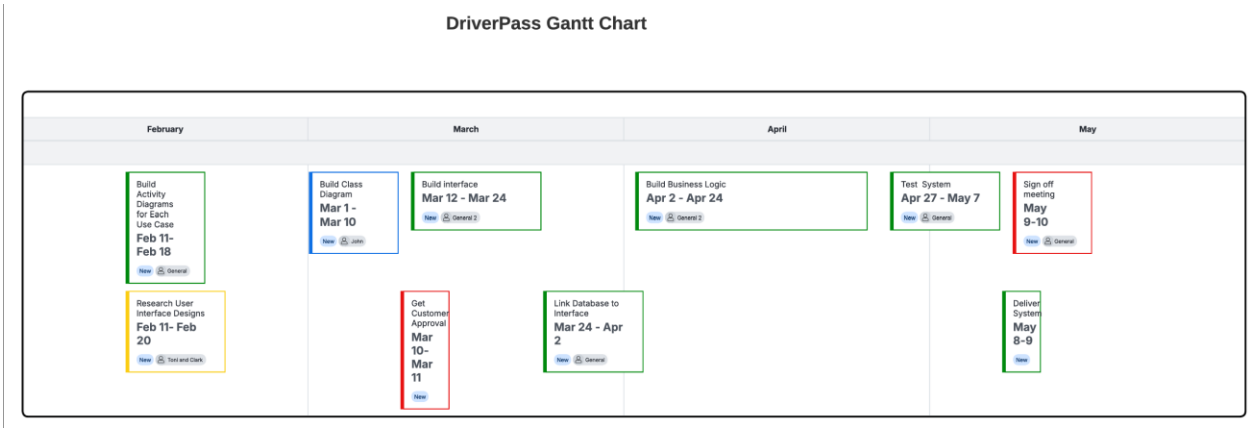
*Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?*

- Financial constraints
    - The more complex and comprehensive the features the more cost they will put on the system, features must be weighed according to the value they provide
- Technical complexities
    - The more comprehensive and complex the features the more probabilities that the system will misbehave or malfunction

## Gantt Chart

*Please include a screenshot of the GANTT chart that you created with Lucidchart. Be sure to check that it meets the plan described by the characters in the interview.*

*******DISCLAIMER: THE INTERVIEW TRANSCRIPT AND FOLLOWING DATES CHART DO NOT MATCH UP I DID MY BEST FROM THE INTERVIEW TRANSCRIPT  BUT ITS HARD TO KNOW WEEKDAYS VS WEEKENDS AND ALL THAT ALSO SOME DATES ONLY APPEAR ON THE CHART AFTER THE INTERVEW AND NOT IN THE INTERVIEW ITSELF*******



Works Cited

*Why partnership matters*. (2026, August 20). Amazon Web Services, Inc. https://aws.amazon.com/isv/saas-scaling-and-partnership/

Valacich, J., & George, J. (2017). Modern Systems Analysis & Design (10th ed.). Pearson. https://bookshelf.vitalsource.com/reader/books/9780138180294/epubcfi/6/2[%3Bvnd.vst.idref%3Dcover]!/4/2%4054:46